

Breeding for Tomorrow Strategic MELIA Framework – empowering the performance agenda

A strategic path forward for developing the Breeding for Tomorrow Science Program's MELIA framework between June- October 2025.

Note to ISDC readers: Because of the size and complexity of the Breeding for Tomorrow Program, we are still finalizing the MELIA plan. We expect this to be ready by end August. In this document we outline how we intend to approach the MELIA plan and what the next steps are for finalization.

Table of Contents

1. Purpose of this document.....	4
2. Program/Accelerator overview and results framework (evaluand overview)	5
3. B4T's MELIA vision	7
Evaluation criteria	7
4. Program/Accelerator overview and results framework	12
B4T Narrative and TOC.....	12
B4T Program Logic	16
Research questions and their expected use	29
B4T Program Logic research questions, and indicators	32
5. Future work required to complete MELIA planning	46
6. Appendices.....	48
Feedback on proposal TOCs and recommendations from the Genetic Innovation Science Group evaluation	48

Glossary

Term or abbreviation	Definition
AOW	Area of Work
B4T	Breeding for Tomorrow
Breeding networks	Networks of CGIAR and NARES formed around crops and regions
Breeding system	The holistic, multidisciplinary breeding innovation ecosystem of stakeholders and partners, breeders and non-breeders
GI	Genetic Innovation
HLO	High Level Output
IA	Impact Assessment
IO	Intermediate Outcome
ISDC	Independent Science for Development Council
KEQ	Key Evaluation Question
MELIA	Monitoring, Evaluation, Learning and Impact Assessment
PCU	Project Coordination Unit
RQ	Research Question
TOC	Theory of Change

1. Purpose of this document

Breeding for Tomorrow (B4T) is a major investment into the genetic improvement of food production systems by CGIAR and partners' breeding networks. The entire MELIA plan for B4T could not be fully finalized within the Inception Phase and therefore, this document outlines those parts of B4T's MELIA that are concrete, as well as detailing a strategic MELIA framework for B4T that:

- Responds to feedback from the ISDC on the proposal TOC and MELIA (see Appendix 1 for further details).
- Takes the opportunity to improve the B4T approach to MELIA, by establishing a robust and logical TOC as a framework for connecting all aspects of MELIA, including the use of tools that have been developed under the previous Genetic Innovation (GI) initiatives and tools that are planned to be developed in B4T.
- Documents progress that has occurred since the 2024 proposal development, particularly on indicators assigned to HLOs.
- Presents a cohesive and strategic TOC and MELIA framework that can provide the basis for applied MELIA in B4T.

The logical framework presented will build an understanding of how MELIA can make use of existing performance management tools developed during the Genetic Innovation Science Group (GI) for informative evaluation. This document is structured in alignment with the guidance provided by CGIAR but tailored to the current stage of MELIA development within the complexities of B4T. The next steps are documented in the section titled 'Future work required to complete MELIA planning'.

2. Program/Accelerator overview and results framework (evaluand overview)

B4T is organized into five Areas of Work (AOW) and 22 High Level Outputs (HLOs) (see Table 1). Each HLO links to an AOW and Intermediate Outcome (IO).

Table 1 Summary of each AOW and the HLOs

AOW	HLO name	HLO summary
1. Market Intelligence (MI): Development of market intelligence, packaged in user-appropriate tools, to support breeders and other innovators make investments with the greatest potential for impact.	HLO1.AOW1.IO1. Steer to impact	Market intelligence is packaged into decision-support tools to guide investment decisions in breeding and other programs
	HLO2.AOW1.IO1 Target markets	Market intelligence is gathered to evidence demand-led and foresight opportunities for impact, organized into market segments
	HLO3.AOW1.IO1 Design concepts	New product concepts are designed for demand-led and foresight opportunities, for use in a Product Design Standard.
	HLO4.AOW1.IO1. Foster motivations	Intelligence gathered on motivations (social, economic, behavioral) for change across the supply chain, consumers and smallholder farmers
	HLO5.AOW1.IO3. Investment cases	Investment case studies are developed to inform investments for maximized impact (change in targeted SDGs)
2. Accelerated Breeding (AB): Creation of products from strategic design of pipelines, discovery and collaborative development of scaling ready products.	HLO6.AOW2.IO1. Refocus to markets	Feasible breeding pipelines are aligned to market segments and opportunities, as guided by product concepts.
	HLO7.AOW2.IO2. Accelerating breeding schemes	Design of optimized breeding schemes that maximize the potential for genetic gain.
	HLO8.AOW2.IO2. Discover genetic potential	Discovery and development of foundational genetic resources such as traits, high-value haplotypes and elite lines for TPPs.
	HLO9.AOW2.IO2. Transform collaboration	Partners are embedded in decision-making for optimized breeding schemes
	HLO10.AOW2.IO2. Create products	Breeding networks collaborate on the production of targeted products.
3. Inclusive Delivery (ID): Development of tools, resources and policies that assist scaling partners be more	HLO11.AOW3.IO4 Position for demand	Strategies to position quality seed and increase scaling partner demand for new varieties, encouraging variety replacement and Product Lifecycle Management.
	HLO12.AOW3.IO5 Deploy seed	Technologies and strategies to encourage scaling partners to increase production and equitable supply of new varieties.

AOW	HLO name	HLO summary
efficient and effective, so that products reach intended markets	HLO13.AOW3.IO5 Power seed scaling	Technologies, strategies and regulatory solutions to encourage scaling partners to accelerate delivery of new varieties to smallholders. Note a more apt name may be 'Fast seed'
	HLO14.AOW3.IO5 Track seed	Methods and tools to track variety adoption and turnover at scale
4. Breeding Resources (BR): Access to tools and services for modernized breeding	HLO15.AOW4.IO2 Hub & spoke scaling	Interoperable tools and services scaled to match the varied capacities of partners across the breeding network, with high-capacity partners enabled to act as central hubs, providing in breeding network support to less resourced partners.
	HLO16.AOW4.IO3 Shared services	A portfolio of connected tools and services that collectively meet current and emerging needs for modernized breeding.
	HLO17.AOW4.IO3. FAIR principles	All breeding programs are supported to maximize alignment with FAIR principles
5. Enable (E): Enabling organizational efficiencies and effectiveness through evidence based active continuous improvement	HLO18.AOW5.IO2. Partnered networks	Strategic planning of crop x region breeding networks enabling collaboration on common challenges and opportunities.
	HLO19.AOW5.IO2 Share capacity	Coordinated capacity-sharing for breeding networks, enabling a dynamic future workforce with appropriate infrastructure and resources in CGIAR and NARES breeding partners.
	HLO20.AOW5.IO3. Assess performance	Evaluation and assessment of performance data gathered by AOWs, and any additional data gathering required to address gaps in the MELIA framework
	HLO21.AOW5.IO3 Improve management	Learnings and bottlenecks identified in Assess performance are addressed through organizational restructuring, enabling maximum impact from (return on) investment
	HLO22.AOW5.IO5. Exchange genetics	Recommendations and guidance on genetic resources policies and practices relating to the use and exchange of genetic resources

3. B4T's MELIA vision

The vision for B4T's MELIA is for an integrated framework that connects beyond the breeding pipeline into an innovation ecosystem with an economic backing which caters to and understands market needs, the genetics of breeding, and the social and policy science of seed delivery systems.

The vision is for a unifying set of nested TOCs that provide a robust basis for a MELIA framework, encouraging collaboration across the AOWs and making use of applied breeding performance management tools (many previously developed in GI) to gather breeding specific evidence to inform evaluation and impact assessment. Significant effort has been invested to ensure the research questions are tightly linked into the purpose of the evaluation and address the evaluation criteria. The research questions provide clear guidance to B4T on the MELIA knowledge required. The research questions are designed to direct studies to information that will improve B4T, enabling second and third loop learning.

Evaluation criteria

This section defines B4T's MELIA objectives for B4T and associated key evaluation questions, before identifying relevant supporting research questions, as guided by the CGIAR evaluation criteria.

The MELIA objectives for B4T are:

- Assess contribution to impact to inform future investment decisions.
- Provide accountability by demonstrating how effective the Program was at achieving its intermediate outcomes.
- Actively learn for continuous improvement in Program design, coordination and implementation (AOW result level).

These objectives have been translated into Key Evaluation Questions (KEQs):

- KEQ1. How has B4T's contribution to the impact areas (return on investment) been maximized?
- KEQ2. To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes?
- KEQ3. How can AOWs influence on intermediate outcomes be increased?

CGIAR has identified seven evaluation criteria. In addition, a criterion of learning has been included to ensure that MELIA actively and explicitly looks for and informs improvements beyond the usual recommendations expected from any MELIA study. The evaluation criteria have been applied in Table 2 to ensure that design thinking in this high-level MELIA framework is comprehensive. A level of importance is provided for each evaluation criteria with a brief explanation. The key evaluation questions and supporting research questions are then defined. The research questions will guide more focused studies to address the Key Evaluation Questions. A lead AOW has been allocated to each research question.

Finally, in the AOW level program logic tables (Table 3) where the assumptions are defined, the research questions are also shown. This provides an additional check on the design integrity of the MELIA framework, ensuring each part logically fits together and details have not been overlooked.

Table 2 Assessment of evaluation criteria, plus learning and resulting key evaluation and supporting research questions

Evaluation criteria	Rating	Rationale	Key evaluation questions (KEQ)	Supporting research questions
Relevance (objectives and program design respond to needs)	High	The prioritization of products has been of increasing interest to investors. It has a significant effect on the impact (return) achieved. Relevance for B4T is primarily concerned with the selection and development of TPPs. The supporting research questions must be addressed to answer the key evaluation question on impact. The criteria of relevance is a critical underpinning to answer impact.	Research questions will contribute to impact KEQ 1. How has B4T's contribution to the impact areas (return on investment) been maximized?	RQ1 To what extent can impact be maximized, without variety dominance that reduces system diversity and resilience? (AOW1 MI) RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products (AOW2 AB)? RQ28 To what extent is market intelligence used?
Coherence (compatibility of the program with other interventions ; synergies and linkages)	Medium	There are three scales of coherence – within regions, along the breeding pipeline, and within CGIAR. Recent work on the development of partnerships and network strategies will continue in B4T. Network strategies are an opportunity to improve coherence with regional and country level policy and investments. Achievement of impact requires products to progress through seed delivery to reach targeted smallholders. Coherence between the breeding pipeline and seed system is a critical keystone step.	Research questions will contribute to KEQ2 on effectiveness: To what extent has B4T and its AOW effectively achieved the intended intermediate outcomes?	RQ4 How have the network strategies improved coherence with region x crop investments? (AOW5 E) RQ5 How have bottlenecks or barriers between variety development and seed systems been overcome to improve smallholder access to seed? (AOW3 ID) RQ6 How has B4T's work with other CGIAR investments increased or improved the achievement of 2030 outcomes? (AOW5 E)
Effectiveness (the extent that the intended results or outcomes are achieved)	High	This criterion directly relates to the MELIA objectives. Assessing the effectiveness of the Intermediate Outcomes will inform contribution analysis and inform the design requirements of future programs.	KEQ2. To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes?	RQ2 How did prioritization using the Design Standard contribute to maximized impact (AOW1 MI)? RQ7 To what extent has B4T increased uptake of varieties by scaling partners who have a widespread reach (collectively)? (AOW3 ID) RQ8 To what extent has CGIAR-NARES prioritization of products encouraged targeted investment for breeding? (AOW2 AB)

Evaluation criteria	Rating	Rationale	Key evaluation questions (KEQ)	Supporting research questions
				RQ9 To what extent has the amount and quality of active collaboration on products increased? (AOW5 E) RQ11 To what extent has B4T influenced more equitable access to improved varieties by marginalized groups? (AOW3 ID) RQ27 To what extent have interoperable tools and services enabled new and improved collaborations? (AOW4 BR) RQ3 will also contribute to effectiveness
Efficiency (delivers results or outcomes in an economical and timely way)	Low	Efficiency gains in breeding contribute to increased rate of genetic gain, as captured in the extent of improvement and impact achieved by products. Efficiency is included in an intermediate outcome so any effectiveness evaluation of intermediate outcomes will be duplicative. This criterion is most relevant to AOW4 as it is designed to support efficient access and use of modernized breeding. An AOW level outcome has captured this so any effectiveness evaluation of outcomes will duplicate efficiency.	Research questions will contribute to KEQ2 on effectiveness: To what extent has B4T and its AOW effectively achieved the intended intermediate outcomes?	RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR) RQ10 To what extent have systemic, organizational changes improved breeding efficiency and effectiveness? (AOW5 E)
Impact (program contribution to impacts)	High	This criterion directly relates to the MELIA objectives. Impact can only be achieved from the collective contributions of all B4T AOWs. Impact is a combination of the extent of improvement in the individual product and the number of smallholder farmers that adopt the product.	KEQ1 How has B4T's contribution to the impact areas (return on investment) been maximized?	The research questions for relevance also apply here. The research questions on learning and effectiveness will assist contribution analysis. RQ13 Have the expected projections from product concept and design translated into on-farm trial results, and if different why/how has more or less been achieved? (AOW2 AB, AOW3 ID)? RQ14 What additional impact could be achieved with additional investment? (AOW1 MI) RQ15 What is the trajectory of genetic gain for each crop, and what benefits could this mean for

Evaluation criteria	Rating	Rationale	Key evaluation questions (KEQ)	Supporting research questions
				smallholder farmers who adopt improved products? (AOW4 BR) RQ16 When is the scale of adoption for products in the pipeline expected to be sufficient to reduce weighted average variety age (WAVA) in targeted market segments to 10 years or less? (AOW4 ID) RQ17 What proportion of marginalized beneficiaries in each market segment have accessed improved varieties? (AOW3 ID)¹ RQ29 What extent of value proposition is needed to motivate adoption?
Sustainability (longevity of benefits after the program finishes)	Low	The long-term benefits of improved crop varieties, once they have reached target beneficiaries, have been previously studied and documented. There is little to be gained from re-investigating this. Sustainability is relevant to three very specific aspects of B4T. In breeding most investment occurs within the pipeline and there could be insights gained from investigating the legacy from influencing seed systems. Capacity building investments should also have legacy for breeding programs to continually improve. There has been increased interest in longer-term planning in replacement strategies, documenting the lifespan of current and improved varieties.	Research questions will contribute to KEQ1 on impact: How has B4T's contribution to the impact areas (return on investment) been maximized?	RQ18 To what extent have capacity improvements been maintained within breeding networks? RQ19 How has the changed approach to networked partnerships contributed to more sustainable breeding networks? (AOW5 E) RQ20 How can long-term replacement strategy planning support scaling partners to prepare for optimum lifespans of developed varieties? (AOW3 ID)
Quality of Science (scientific credibility and legitimacy)	Medium	High quality breeding uses modern techniques to successfully achieve the desired target product profile, with substantial genetic gain. There is low error, with the influence of environmental conditions controlled and high heritability of targeted traits. Quality of science is required for	Research questions will contribute to KEQ2 on effectiveness: To what extent has B4T and its AOW effectively achieved the intended	The first part of RQ15 What is the trajectory of genetic gain for each crop, is relevant here. RQ21 To what extent have market set targets been successfully converted into breeding targets? (AOW2 AB)

¹ It would be ideal to question if the level of benefit gained from adoption of improved varieties was sufficient for empowerment. However, determining the sufficiency of benefit is likely to be unfeasible in the timeframe and resourcing available. Opportunities to incorporate data collection at this depth of enquiry should be kept in mind.

Evaluation criteria	Rating	Rationale	Key evaluation questions (KEQ)	Supporting research questions
		breeding to be successful and achieve impact. This quality, combined with critical mass of partnerships and corporate knowledge and experience, that supports ongoing competitive advantage.	intermediate outcomes (in particular successful product development)?	RQ22 How has the scaled use of modernized practices contributed to maintaining CGIAR's competitive advantage in breeding? (AOW4 BR) RQ23 What contribution have game-changing genetic resources made to impact from variety development? (AOW2 AB)
Learning	High	This criterion directly relates to the MELIA objectives. Each AOW will need to perform and deliver results in order for B4T to achieve the intermediate outcomes and impacts. In addition, Enable will provide MELIA leadership and across-AOW rigour to learning generation.	KEQ3 How can AOWs influence on intermediate outcomes be increased?	RQ24 Have the AOWs effectively come together in transdisciplinary collaboration that holistically encompasses breeding from market targeting through to delivery to smallholders? (AOW5 E) RQ25 What organizational changes are needed to overcome challenges or increase return (impact) on investment? (AOW5 E)

4. Program/Accelerator overview and results framework

B4T Narrative and TOC

Internal thinking assisted with clarity on the purpose of each AOW, along with how the AOWs will come together to deliver impact. The internal AOW results have helped to improve assumptions, providing more logical causal pathways between HLOs and Intermediate outcomes. The revision of causal pathways has resulted in refinement to the Intermediate and 2030 Outcomes. The HLO names have been slightly expanded to provide greater clarity on what each HLO is seeking to achieve. As a result, the TOC narrative now reads as follows.

Market Intelligence (AOW1) will develop knowledge on demand-led and foresight opportunities for impact. The knowledge will be packaged into decision-support tools (HLO1. Steer to impact). Market segments will be defined (HLO2. Target to market) and combined with the decision-support tools, will be used to develop product concepts (HLO3. Design concepts). Combined these HLOs will develop knowledge for breeding networks to apply, informing the design of impact- and demand-oriented products.

Accelerated Breeding (AOW2) will interpret and apply market intelligence and develop technically feasible strategies for Target Product Profiles (TPPs) and pipelines (HLO6. Refocus to markets). CGIAR-NARES breeding networks will then be able to prioritize the transdisciplinary designed, high impact products (HLO7. Accelerating genetic gain). Joint prioritization will encourage more targeted breeding by networks (IO1).

The cost-effectiveness of the breeding network (IO3) will be achieved through

- A culture of continuous improvement will be led by Enable (AOW5) as it works to address common challenges across the breeding networks. Breeding performance will be monitored annually, with rectifying improvements to organizational structures and processes (HLO21 Improve management), overcoming bottlenecks such as policy barriers to the exchange of genetic resources (HLO22 Exchange genetics).
- Market Intelligence (AOW1) will develop investment cases that forecast potential benefits of breeding investments, and the results will be used as a check on the level of impact that is emerging or has been achieved. Analysis of forecasts compared with actual results gained from example, in-depth impact assessments by Enable (AOW5) (HLO20 Assess performance), will test scaling assumptions, providing additional confidence in the projection models. Combined these activities will ensure B4T remains focused on delivering maximum impact.
- Breeding Resources (AOW4) will provide shared services, with economies of scale used to increase the cost-effectiveness of breeding tools and services, to make modernized breeding more available and feasible through the CGIAR-NARES breeding network.

The culture of continuous improvement, enabled through active improvements, maintained impact-orientation and shared services will come together in CGIAR-NARES breeding networks that efficiently develop products, in the shortest possible time (IO3). The improved processes and modernized approaches will further enhance the success of breeding activities (IO2).

Active collaboration to successfully develop products (IO2) will occur as a result of HLOs across three AOWs.

- Accelerated Breeding (AOW2) will discover and develop foundational genetic resources such as traits, high-value haplotypes and elite lines (HLO8. Discover genetic potential), resulting in traits and genetic resources with significant potential for genetic gain.
- Enable (AOW5) will develop new approaches to partnerships with strategic planning of crop x region breeding networks, enabling collaboration on common challenges and

opportunities (HLO18 Partnered networks), and coordinate capacity-sharing that enables a dynamic future workforce with appropriate infrastructure and resources (HLO19 Share capacity). Combined the multi-scaled and multi-disciplinary partnered networks with appropriate capacity will be enabled to implement modernized breeding.

- Accelerated Breeding (AOW2) will build on the networked partnerships established by AO5: Enable (HLO18 Partnered networks) and embed partners into decision-making on breeding schemes and pipelines (HLO9 Transform collaboration).
- Breeding Resources (AOW4) will develop a package of interoperable tools and services that are scaled to align with the varied capacity within breeding networks. More resourced and higher-capacity partners will be enabled to act as central hubs, providing support to less resourced partners. This approach will build collaborative interdependence and more resilient networks.

The development of genetic resources with significant potential for genetic gain, combined with partnered networks capable of modernized breeding, resourced with interoperable scaled tools, will come together in CGIAR-NARES breeding for targeted products (HLO10 Create products), in accordance with the product prioritization from IO1. This will result in CGIAR Centers, AOWs and NARES actively collaborating on successful product development (IO2).

Successful product development that delivers the targeted levels of genetic gain and traits required for impact-led and demand-led products will encourage farmers to adopt new varieties.

Inclusive delivery (AOW3) will work on both the supply and demand side of variety adoption. Insights on the motivations of scaling partners and smallholder farmers (HLO4 Foster motivations) will provide feedback to Market Intelligence and inform replacement strategies that along with on-farm trials that demonstrate performance, will increase demand for new varieties by scaling partners (HLO11 Position for demand). Increased demand for improved varieties will further increase uptake by smallholder farmers (2030-OC1)

On the supply side, technologies and strategies to encourage scaling partners to increase production and equitably distribute seed (HLO12 Deploy seed), and technologies and strategies to accelerate delivery (HLO13 Power seed scaling) will address both volume and speed limitations to improve smallholder access to improved varieties. Combined with policies on the exchange of genetic resources (HLO22 Exchange genetics), increased supply capacities will lead to NARES and SMEs promoting equitable access of improved varieties by marginalized groups (IO5). Use of improved varieties by marginalized groups will empower them (2030-OC2).

Significantly improved products with widespread use by smallholder farmers and marginalized groups, will multiply individual benefits to create a transformative, societal scale of benefits for farmers, communities and society.

There are now multiple pathways to impact, with adoption maximized through a combination of collaboration for greater NARES empowerment and ownership of results, enhancement of supply and demand by scaling partners, rigorous prioritization of impact- and demand-oriented products with associated resource targeting and the use of modernized and efficient breeding practice that ensures successful development of step-change products. In this comprehensive strategy the breeding pipeline is encompassed within a holistic innovation ecosystem, that uses market intelligence and inclusive delivery to achieve societal scale impacts.

Assumptions

The full list of assumptions, identified for each component of the TOC are detailed in the program logic table (see Table 3). Assumptions can relate to a causal link, or details within an outcome statement. The TOC Board only allows for causal assumptions, and these need to be classified as one of:

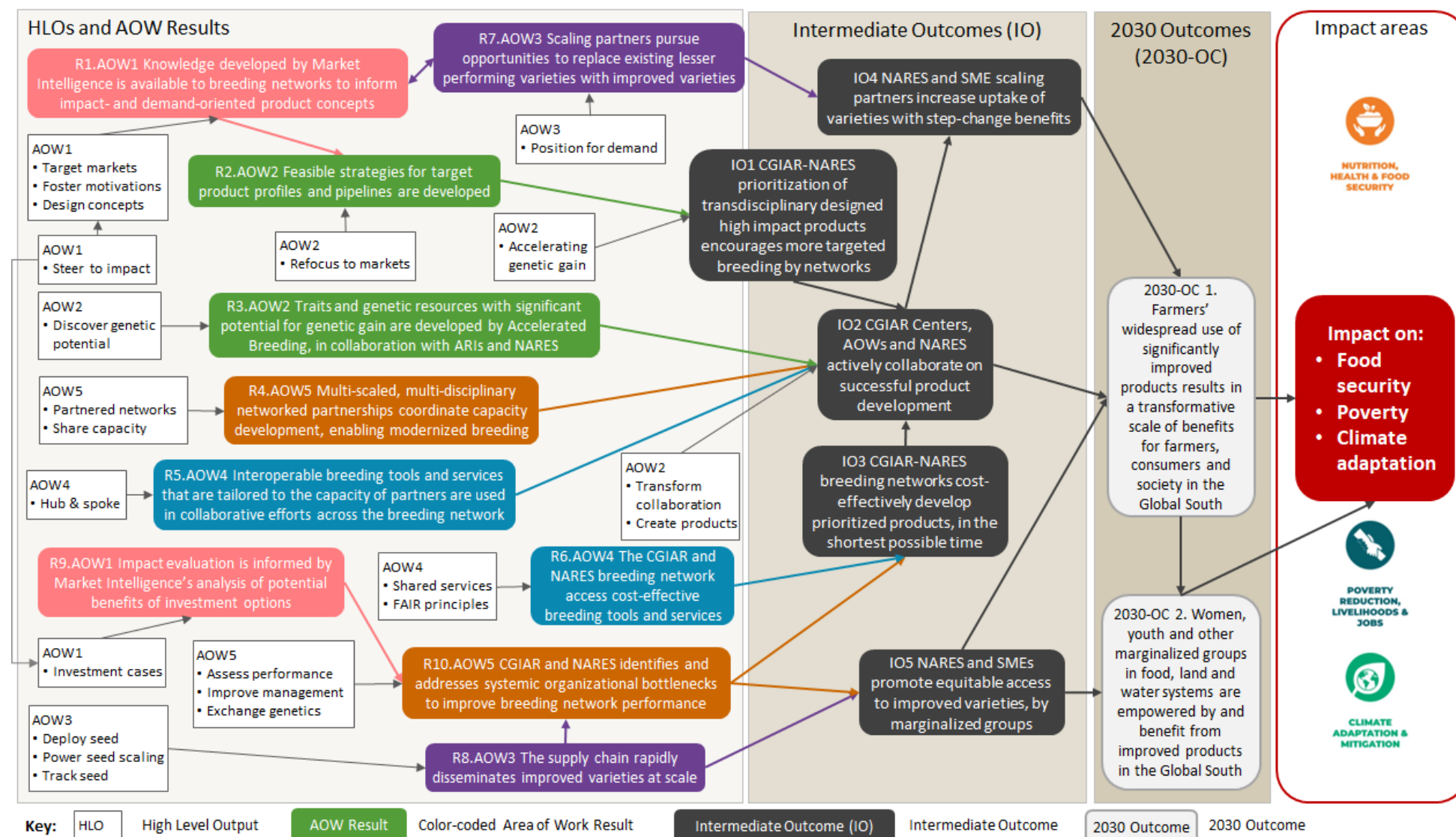
- Actor is convinced to act
- Actor is empowered to act
- New or improved knowledge/technology makes it possible to act
- New or improved relationships make it possible to act
- New or improved conditions make it possible to act.

This has prevented a number of assumptions from being incorporated into the TOC Board, particularly those that relate to innovation development within a HLO or Outcome (i.e. not related to a causal pathway). For example, the assumption ‘Large step-change levels of improvement are technically possible within the political limitations on breeding technologies’ is not about a connection between TOC parts, but a context related to the HLO Discover genetic potential.

When only causal assumptions are considered, and assumptions inform identification of relevant Research Questions, this can be limiting to MELIA studies. There are a handful of major assumptions that warrant some thought. A couple of these assumptions require some high-level strategic consideration as the assumption highlights potential discord in CGIAR directions on legacy, resilience and impact.

- There is no trade-off between transformational impact at scale and diversified climate resilient farming systems.
- The accumulation of individual benefits leads to a transformative scale of benefits.
- There is sufficient commonality of needs and market segments across the breeding networks for partners to have shared, distributed benefits from the prioritized products, encouraging convergence and collaboration rather than coordinated independence.
- Prioritization of products is compatible with a desire for systems resilience through crop and variety diversity.
- Resource limitation is an allocation issue, rather than a volume issue (i.e. prioritization and reallocation of resources will enable successful product development).
- Scaling interventions do not erode competitive advantage or branding/market differentiation of individual seed suppliers or provide unfair advantages and distort seed markets.
- Breeders have the social science skills to correctly interpret and apply social science data without direct involvement or support of social scientists.
- There is no difference in the ability of NARES and CGIAR breeders to interpret and apply social science, with no power imbalance created in joint decision-making processes due to social science capacity differences.
- Feasibility of product concepts applied before prioritization does not block or limit the innovation required to achieve transformative levels of change in product performance.
- Coherence (of CGIAR and partner priorities) does not have negative effects on farming system resilience.
- Product performance does not provide a sufficient valuable proposition for scaling partners, requiring interventions to motivate uptake.
- Legacy, as the long-term sustained impacts from an innovation, is compatible with climate change requirements for variety turnover every 10 years.

Figure 1 TOC for B4T, with AOW results and HLOs



B4T Program Logic

A simplified program logic table has been developed to ensure alignment of all MELIA components and activities to the one common TOC, so that results and findings will come together in a cohesive and comprehensive whole.

Table 3 - Program logic for B4T

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
2030-OC1 Farmers' widespread use of significantly improved products results in a transformative scale of benefits for farmers, consumers and society in the Global South	Improved products are demand-led and impact-focused, providing a compelling value proposition for smallholders and consumers. This means that improved is a generic term that encompasses all advantages trait changes, beyond yield. Products are adopted by a large proportion of the targeted market segments. The combination of product improvement and extent of uptake, creates a transformative scale of benefits.	Improved varieties provide a large-enough gain in value proposition for smallholder farmers that it motivates individual farmers to replace varieties that are over 10 years old. <i>Note HLO4.AOW1 Foster motivations will quantify 'large-enough'.</i> Provisionally, a productivity gain of 30% is considered transformational for an individual farmer. There is a large number of farmers taking up improved products, with adoption replacing the market share of older varieties. Provisionally, a market share between 25% and 50% is a large enough proportion to have significant impact, while ensuring some variety diversity and resilience remains. <i>Note HLO2.AOW1. Target markets will need to determine optimal levels of variety uptake.</i> The extent of improvement in the variety, combined with the number of farmers using the variety results in a large total benefit in targeted market segments that is detectable in a	KEQ1: How has B4T's contribution to the impact areas (return on investment) been maximized? RQ13 Have the expected projections from product concept and design translated into on ground results, and if not, why not (AOW2 AB, AOW3 ID)? RQ14 What additional impact could be achieved with additional investment? (AOW1 MI) RQ15 What is the trajectory of genetic gain for each crop, and what benefits could this mean for smallholder farmers who adopt improved products? (AOW4 BR) RQ16 When is the scale of adoption for recently released varieties and products in the pipeline expected to be sufficient to reduce weighted average variety age (WAVA) in targeted market segments to 10 years or less? (AOW4 ID)	There is no trade-off between transformational impact at scale and diversified climate resilient farming systems. A correlation can be detected between global measures and variety adoption (i.e. absence of other mitigating factors). Products that meet smallholder and consumer needs will be enough to motivate widespread adoption. The accumulation of individual benefits leads to a transformative scale of benefits. Seed scaling limitations can be overcome, so availability and access to seed is no longer a barrier to adoption.

² Some improvement in processes or data may be needed to achieve these results. Rather than limiting to the current state, a forward-looking view provides guidance and motivation to areas that may require continuous improvement. E.g. detailed cost monitoring during breeding implementation to inform budgeting and financial recovery models.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
		correlating shift in trajectory of targeted global measures in the market segment.	RQ6 How has B4T's work with other CGIAR investments increased or improved the achievement of 2030 outcomes? (AOW5 E)	
2030-OC2 Women, youth and other marginalized groups in food, land and water systems are empowered by and benefit from improved products in the Global South	Products are designed and developed to address the unique needs of marginalized groups. Marginalized groups are able to access these products and benefit from them.	A large proportion of marginalized groups gain access to seeds of improved varieties. The use of these varieties, and benefits that ensue, are sufficient to empower marginalized groups and reduce their level of disadvantage. The proportion of marginalized product users in each market segment is known, and over 50% of these users access varieties that are designed specifically for them.	RQ17 What proportion of marginalized beneficiaries in each targeted market segment have accessed improved varieties? (AOW3 ID)	Access to improved varieties will have sufficient benefits to reduce relative disadvantage experienced by marginalized groups. Marginalized groups have unique product needs. Product needs of marginalized groups are compatible with demand-led and foresight products. Products targeted for marginalized groups deliver sufficient impact to be competitive for investment against products that have broader targets and potential for impacts to be experienced by a larger number of beneficiaries. Wrap around services or support is not required for marginalized groups to maximize the benefits gained from the new varieties. Marginalized groups are distributed across market segments and locations serviced by the breeding network, limiting the emergence of 'freeloader' issues if less resourced partners benefit or benefit more from products targeted to marginalized groups, with lesser relative investment in the collaborative breeding efforts.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
IO1 CGIAR-NARES prioritization of transdisciplinary designed high impact products encourages more targeted breeding by networks	Partners systemically contribute to innovation, priority-setting, decision-making and the development and delivery of farmer-valued cultivars. The contribution of NARES partners makes a difference to breeding prioritization. I.e. not a ratification of presented options. The collaborative approach encourages partners to reallocate resources to prioritized products. The reallocation of resources means that breeding efforts increase their chances of success. The Product Design Standard is applied and ensures appropriate prioritization.	The breeding network competently uses social science data to inform product prioritization, and this use of social science data makes the prioritization process transdisciplinary. The inclusion of social science in the Product Design Standard is sufficient to ensure decision-making overcomes existing breeding biases. The prioritized products aim to deliver (provisionally) at least 80% of the averaged projected benefits across the product concepts used to inform product designs. Across the CGIAR-NARES breeding network resources are reallocated to agreed priorities, while maintaining distributed ownership and empowerment across the network. Partners are committed to the prioritized products. The extent of reallocation is sufficient to ensure prioritized products are successfully developed within the shortest possible timeframe.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ8 To what extent has CGIAR-NARES prioritization of products encouraged targeted investment for breeding? (AOW2 AB)	There is sufficient commonality of needs and market segments across the breeding networks for partners to have shared, distributed benefits from the prioritized products, encouraging convergence and collaboration rather than coordinated independence. Prioritization of products is compatible with a desire for systems resilience through crop and variety diversity. Partners will be motivated to reallocate resources to the network priorities. The reallocation of resources to priorities is not disadvantages to individual partners. The work prior to prioritization does not limit or remove choices that would have been more attractive to NARES partners. Product designs can be coordinated into breeding schemes and pipelines without losing sight of the market determined targeted level of value proposition required to motivate adoption and deliver impact.
IO2 CGIAR Centers, AOWs and NARES actively collaborate on successful product development	Breeding networks successfully produce candidate products for major food, food-feed, and forage crops, vegetables and trees with the ability to drive a	Successful product development means that the scale of improvement targeted, as defined to deliver transformative impact, is achieved. Active collaboration is where knowledge sharing prompts discussion in a process of curiosity and discovery. This contrasts with passive	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ9 To what extent has the amount and quality of active collaboration on products increased? (AOW5 E)	Success is enabled by prioritization and shift in resources. Power imbalances between participants in active collaboration do not restrict the ability for different voices, perspectives and areas of expertise to be heard, either within active collaboration forums/spaces,

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
	transformative scale of impact.	collaboration where knowledge is provided or gained, without coming together to explore differing experiences or perspectives. Active collaboration occurs across organizations, scales and work units.		or afterwards in organization workplans, processes and decisions. The ability to successfully develop new products is improved as a result of active collaboration.
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ24 Have the AOWs effectively come together in transdisciplinary collaboration that holistically encompasses breeding from market targeting through to delivery to smallholders? (AOW5 E)	There is the required capacity (skills, culture etc) to embrace diverse views and jointly address challenges using transdisciplinary participation. Budget limitations and inherent power imbalances do not create protectionist attitudes that marginalize adjunctive areas of expertise.
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products? (AOW2 AB)?	The reallocation of resources as a result of prioritization is sufficient for successful product development. Resourcing is currently limiting or preventing successful product development, including the extent of improvement in new products (e.g. genetic gain). Additional resources will overcome this. Resource limitation is an allocation issue, rather than a volume issue. There is sufficient capacity within CGIAR, ARIs and NARES for joint-problem solving efforts, without undermining power imbalances.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ27 To what extent have interoperable tools and services enabled new and improved collaborations? (AOW4 BR)	Incompatible tools and services are creating barriers to collaboration, such as Multi-Environment Trials conducted across locations and organizations.
IO3 CGIAR-NARES breeding network cost-effectively develops prioritized products, in the shortest possible time	Efficiency, in terms of cost-effectiveness, is improved as result of access to modernized tools and services and organizational bottlenecks. These changes reduce the amount of time it takes to develop a new product from concept to deployment.	Breeding efficiency is more than just the cost of breeding. Efficiency is the cost of successful product development (i.e. cost-effectiveness). This interpretation combines efficiency from IO3 with successful in IO2, ensuring compatibility of these outcomes. Ensuring successful breeding may require increased cost where previous investment has been insufficient. Products prioritized based on impact may be more expensive to develop. The shortest possible time varies between crops due to the biological growth cycle.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ10 To what extent have systemic, organizational changes improved breeding efficiency and effectiveness? (AOW5 E)	There are common systemic challenges and bottlenecks across the breeding networks that can be resolved at a larger scale. Organization specific contexts do not prevent single at scale solutions.
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR)	Economies of scale apply to the use of breeding tools and services. CGIAR is able to use its scale of operations to negotiate improved pricing. Centralized services do not excessively increase administrative burdens, beyond the efficiency savings gained. Common use of tools and services enables interoperability and data sharing across partners. Data sharing improves the cost-effectiveness of breeding. Partners have the resources and capacity required to effectively use modernized tools and services.
IO4 NARES and SME scaling partners increase	Scaling partners are motivated to uptake new varieties and make these	The market share of engaged scaling partners collectively covers a large enough portion of the targeted farmers	KEQ2: To what extent has B4T and its AOWs effectively	There are sufficient benefits and motivations for scaling partners to change behavior, systems and seed

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
uptake of varieties with step-change benefits	available to farmers in the relevant market segments.	to allow for below 100% adoption rates and still meet impact targets. I.e. 80% of seed market x 50% farmer adoption = 40% adoption. Scaling partners have the information and resources required to position products to the markets and gain sufficient returns to motivate changes to seed supply.	achieved the intended intermediate outcomes? RQ7 To what extent has B4T increased uptake of varieties by scaling partners who have a widespread reach (collectively)? (AOW3 ID)	decisions. In a commercial or semi-commercial context this requires farmers (seed buyers) to expect sufficient gains from investment in a new variety to pay for the seed and still deliver enhanced profit, and for the price received by the seed supplier to cover any additional costs of changing product lines plus a motivating level of profit. (i.e. distributed benefits through the supply chain). The extent of product improvement provides a motivating level of profit through the supply chain and for farmers. Scaling interventions do not erode competitive advantage or branding/market differentiation of individual seed suppliers or provide unfair advantages and distort seed markets.
IO5 NARES and SMEs promote equitable access to improved varieties, by marginalized groups	Women, youth and other marginalized groups are able to access improved varieties. Varieties that have been designed specifically to reduce the disadvantage of these groups reaches the targeted groups.	Equitable access is fair and equal access to improved varieties, ensuring that marginalized groups are not further disadvantaged and not left behind as others take up improved varieties. Fair and equal access is measured proportionately with an equal proportion of marginalized groups in the target market segment access varieties as the proportion of non-marginalized users of improved varieties.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ11 To what extent has B4T influenced more equitable access to improved varieties by marginalized groups? (AOW3 ID)	There are benefits to scaling partners or sufficient motivations from servicing marginalized groups. Equal access to improved varieties is sufficient to empower marginalized groups. Benefits from uptake of improved varieties by non-marginalized groups will flow to marginalized groups through societal scale changes. There are barriers to the sharing of genetic resources with scaling partners.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
R1.AOW1 Knowledge developed by Market Intelligence is available to breeding networks to inform impact- and demand-oriented product concepts. HLO1.AOW.IO1. Steer to impact HLO2.AOW1.IO1 Target markets HLO3.AOW1.IO1 Design concepts HLO4.AOW1.IO1. Foster motivations	Data is gathered for new market segments and new crops. Existing market segment information is maintained, with an additional layer of complexity added to look at the relationship between crops within regions. This will allow the development of product concepts within resilient, multi-crop farming systems. Data is packaged into decision-making tools and product concepts so that breeders can design products for prioritization and investment.	The use of market intelligence data for decision-making requires three conditions to be met: • The quality and completeness of the information available • Accessibility of the information to decision-makers • Application and interpretation of the information within decision-making processes Each of these three conditions needs to be separately monitored for insights on improvement points along the process. The quality and completeness of the information can be monitored internally by Market Intelligence. Accessibility can be inferred from number of times decision-tools are accessed, and the number of unique users (IP addresses) and repeat users. The last two conditions will benefit from some level of feedback from data users. Finally, assessing correct interpretation will require assessment of the decisions made using market intelligence. This can be done by reviewing results from the Product Design Standard and prioritization processes. A comparison of potential (pre decision) and forecast impact (post decision) will be informative of the interpretation of data and highlight other factors that may be coming into play to limit impacts from breeding.	KEQ1: How has B4T's contribution to the impact areas (return on investment) been maximized? RQ1 To what extent can impact be maximized, without variety dominance that reduces system diversity and resilience? (AOW1 MI) RQ28 To what extent is market intelligence used?	Factors that influence impact and demand-led product concepts and design are consistent over the short-term (5 years) and do not need to be monitored for changes on an ongoing basis. Existing market intelligence is sufficient to develop understanding of the relationships between crops to build knowledge on a resilient, multi-crop approach with product concepts contextualized within a package of products for a market segment. The Product Design Tool can address any potential trade-offs between crops within a market segment, enabling resilient farming systems (i.e. a fifth dimension is not required in the tool to consider implications to crop diversity and resilience)
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ2 How did prioritization using the Design Standard contribute to maximized impact (AOW1 MI)? RQ29 What extent of value proposition is needed to motivate adoption?	Breeders have the social science skills to correctly interpret and apply social science data without direct involvement or support of social scientists. There is no difference in the ability of NARES and CGIAR breeders to interpret and apply social science, with no power imbalance created in joint decision-making processes due to social science capacity differences. There is no loss or limit to NARES empowerment due to capacity differences.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
R2.AOW2 Feasible strategies for target product profiles and pipelines are developed HLO6.AOW2.IO1. Refocus to markets	Breeders convert product concepts into product design, based on the available technologies, tools, services and genetic resources. Value propositions that cannot be achieved within existing systems are deemed too ambitious and filtered out, ensuring that Targeted Product Profiles are reasonably achievable.	Feasible means achievable within currently available technologies and tools. The resourcing required to deliver each feasible product profile and pipeline is estimated to ensure the next stage of prioritization for resource allocation results in sufficient condensing of resources to achieve required technical results. Quantitative targets for the level of improvement required for all value propositions are set. The targets are sufficient to motivate uptake by scaling partners and adoption by smallholder farmers, as informed by market intelligence's data.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ21 To what extent have market-set targets been successfully converted into breeding targets? (AOW2 AB)	Feasibility of product concepts applied before prioritization does not block or limit the innovation required to achieve transformative levels of change in product performance. The resourcing required to implement feasible strategies can be costed to a pragmatic level of estimation, ensuring the next step of prioritization does not result in levels of over or under resource reallocation that prevents achievement of required results for individual products, or create wastage from over reallocation.
R3.AOW2 Traits and genetic resources with significant potential for genetic gain are developed by Accelerated Breeding, in collaboration with ARIs and NARES HLO8.AOW2.IO2. Discover genetic potential	The genetic resources that are discovered and developed target traits required for impact- and demand-oriented products. A step-change level of improvement is possible when these resources are used.	The resources are available to breeders and used to develop improved varieties. This requires breeders to have the capacity required to make use of the resources and ability to access them. A step-change level of improvement is (provisionally) a 50% gain or greater in performance of the targeted trait in haplotypes. This level of improvement allows for some loss as traits are brought into existing lines. Collaboration with ARIs and NARES is more than input into decision-making. It includes shared resources, including genetics and constructive joint-problem solving as difficult breeding challenges are embraced and innovations attempted.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ23 What contribution have game-changing genetic resources made to impact from variety development? (AOW2 AB)	Large step-change levels of improvement are technically possible within the political limitations on breeding technologies. ARIs are willing to share technologies and innovations with the CGIAR-NARES breeding networks. Breeders have the capacity and motivation to make use of genetic resources. Barriers on the distribution of genetic resources do not limit access.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
R4.AOW5 Multi-scaled, multi-disciplinary networked partnerships coordinate capacity development, enabling modernized breeding HLO18.AOW5.IO2 Partnered networks HLO19.AOW5.IO2 Share capacity	Breeding networks innovate and implement market-oriented, sustainable partnership models in which partners systemically contribute to innovation, priority-setting, decision-making and the development and delivery of farmer-valued cultivars. Capacity building events and opportunities are coordinated across the AOWs to address the strategically defined capacity needs of partners. Partners are encouraged and facilitated to share capacity directly, creating some independence from CGIAR, further establishing a network of breeding partners (as opposed to centralization around CGIAR centers).	Networked partnerships are dynamic and complex, with several interrelationships. There are a greater number of connections that do not flow through CGIAR centers. Breeding priorities and efforts remain collaborative towards common goals. Network strategies are comprehensive and set the long-term vision for the network, with clear links to the Intermediate Outcomes and Impacts. Network strategies enable forward planning for all partners, with links to replacement strategies so that future capacity requirements can be developed to ensure future breeding priorities can be delivered. Connections between breeding networks enable greater spillover effects, with innovations developed shared across networks. There may be common links or bridges across networks, with networks operating across geographic scales.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ4 How have the network strategies improved coherence with region x crop investments? (AOW5 E)	Network strategies will consider existing national priorities and also provide guidance for future or in-development national strategies. Coherence supports the allocation of resources to common goals, and successful attainment of those goals. Coherence does not have negative effects on farming system resilience. Politics do not limit willingness to align policies across countries, with associated potential loss of differentiation. A healthy network has connections across a number of partners, as opposed to connections radiating out from a central actor.
			KEQ1: How has B4T's contribution to the impact areas (return on investment) been maximized? RQ19 How has the changed approach to networked partnerships contributed to more sustainable breeding networks? (AOW5 E)	
			KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ18 To what extent have capacity improvements been maintained within breeding networks? (AOW5 E)	
R5.AOW4 Interoperable breeding tools and services that are tailored to the capacity of	Breeding tools and services are designed to be accessible and useful for a range of capacity levels, while maintaining interoperability.	A hub and spoke model of scaled tools and services allows partners of varied capacity and resourcing to make valuable contributions to shared breeding goals. The design of scaled tools and services empowers hubs to	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ22 How has the scaled use of modernized practices contributed	Tools and services can be scaled for differing levels of capacity and resourcing and retain interoperability. Interoperability is maintained over time (e.g. as capacity improves and organizations shift to using more

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
partners are used in collaborative efforts across the breeding network HLO15.AOW4.IO2 Hub & spoke services	Interoperability occurs across the scales or levels of a tool, and between tools. The common use of a consistent set of tools and services helps partners to connect and share data and resources. The capacity of less resourced partners is increased.	work directly with spokes, establishing a sustainable model of respectful, regionalized capacity building.	to maintaining CGIAR's competitive advantage in breeding? (AOW4 BR)	advanced versions of tools and services) and between organizations. Interoperability is a requirement for collaboration. The joint research with common use of data by CGIAR and NARES effectively creates a larger distributed research organization with greater competitive advantage. More resourced and higher capacity hubs are willing and able to provide technical support to lesser resourced spokes. Network independence does not reduce global scale collaborations.
R6.AOW4 The CGIAR and NARES breeding network access cost-effective breeding tools and services HLO16.AOW4.IO3 Shared services HLO17.AOW4.IO3 FAIR principles	Breeding programs access and use the tools and services centrally available to implement modernized breeding. FAIR principles are embedded in shared services and tools.	A critical mass of tool and service users ensures that a minimum operating volume of use is met. The volume of users maintains and improves negotiating power with commercial providers of tools and services. The quality of the tools and services developed by CGIAR are attractive to commercial breeders, generating interest and additional users. Modernized breeding practices (e.g. quantitative genetics) that accelerate and improve genetic gain are routinely used across the majority of the breeding pipelines.	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR)	Access to tools and services is a barrier to the use of modernized breeding tools. Improved access is sufficient to motivated adoption of tools and services. Breeding programs are able to fund upfront costs of tools and services, and these costs are sufficiently offset by labor or time savings from accelerated breeding. Minimum operating volumes to retain economies of scale can be met within the budget constraints of breeding programs. There is the required capacity, or capacity support, to enable use of tools and services, and the results gained.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
R7.AOW3 Scaling partners pursue opportunities to replace existing lesser performing varieties with improved varieties HLO11.AOW3.IO4 Position for demand	Institutional or system barriers to the uptake of improved varieties by scaling partners are overcome. Scaling partners are motivated, informed on product performance and able to make variety scaling decisions that align with and contribute towards replacement strategies.	All policy and institutional barriers have been overcome. The only remaining barrier to the uptake of improved varieties by scaling partners is insufficient expected benefits or returns of improved varieties. This is the returns from the perspective of the scaling partner i.e. the market forces that flow through from smallholders due to performance gains of improved varieties relative to any scaling costs of changing varieties. Scaling partners have improved understanding of the benefits of changing varieties and are motivated to make use of improved varieties.	KEQ1: How has B4T's contribution to impact areas (return on investment) been maximized? RQ20 How can long-term replacement strategy planning support scaling partners to prepare for optimum lifespans of developed varieties? (AOW3 ID)	There are clear advantages for scaling partners from improved varieties. Product performance does not provide a sufficient valuable proposition for scaling partners, requiring interventions to motivate uptake. There is high confidence that smallholders will be able to achieve the potential gains in product performance, and able to then repay seed scaling partners or afford ongoing use of seed. There are no agronomic changes required, or agronomic changes do not require adjunctive capacity building support for smallholder farmers in order for benefits of improved varieties to be gained. Legacy, as the long-term sustained impacts from an innovation, is compatible with climate change requirements for variety turnover every 10 years. Scaling partners are willing to retire existing profitable (profitable for scaling partners) varieties from their offerings. Longer-term replacement strategies are useful for scaling partners to prepare and business plan for changing product lines (i.e. scaling partners position themselves to change varieties).

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
R8.AOW3 The supply chain rapidly disseminates improved varieties at scale HLO12.AOW3.IO5 Deploy seed HLO13.AOW3.IO5 Power seed scaling HLO14.AOW3.IO5 Track	Seed production for lesser performing varieties is reduced and improved varieties are produced instead. There is both an increase in the volume of seed produced, and the speed that it is produced.	Adoption of improved varieties is not limited by farmer access to seed. There is a small surplus volume of seed available each year for varieties that are less than 10 years old. (A consistent approach for measuring age of variety is under development. This is expected to take into account delays between CGIAR development and release by national partners).	KEQ2: To what extent has B4T and its AOWs effectively achieved the intended intermediate outcomes? RQ5 How have bottlenecks or barriers between variety development and seed systems been overcome to improve smallholder access to seed? (AOW3 ID)	Adoption rates are currently limited by the volume of seed available and/or speed of seed deployment. Farmers are aware of recently released varieties and able to make informed choices on varieties.
R9.AOW1 Impact evaluation is informed by Market Intelligence's analysis of potential benefits of investment options HLO5.AOW1.IO3. Investment cases	Evidence-based projections of benefit are used to inform development of compelling investment cases. There are feedback loops between the investment cases, breeding achievements and impact assessments. Reconciliation of results and findings are used to improve the processes and assumptions used in each.	The information in these investment cases can be used as a check on the level of impact currently being delivered, and discrepancies can be explored to improve either breeding processes or projections. Monitoring and evaluation provide evidence for assumptions in projection models, increasing confidence in the projections of impact as a reliable impact data point. Projections of benefit can be readily updated as products progress through the breeding pipeline and finally based on results of on-farm trials.	KEQ1: How has B4T's contribution to impact areas (return on investment) been maximized? RQ14 What additional impact could be achieved with additional investment? (AOW1 MI)	Projection of benefits using late trial or on-farm trial data will be accepted by Funders and other stakeholders as cost-effective evidence of impact. Data collected through the breeding process and assumptions in projection models are detailed enough to support progressively updated projections and enable investigations to understand differences.
R10.AOW5 CGIAR and NARES identifies and addresses systemic organizational	Processes that are common across breeding networks are refined to reduce organizational bottlenecks and improve	Organizational change supports the widespread adoption of modernized breeding, with a high quality of science and cost-effectiveness. Cultural changes continue with continuous improvement embedded	KEQ3: How can AOWs influence on intermediate outcomes be increased? RQ25 What organizational changes are needed to overcome	There are current bottlenecks impacting breeding network performance, and these bottlenecks are common across organizations and breeding programs.

Outcome, Result or HLO	Explanatory statement	Definition of success (in 5 years ²)	Key Evaluation Question (KEQ) and Research Questions (RQ)	Assumptions
bottlenecks to improve breeding network performance HLO20.AOW5.IO3 Assess performance HLO21.AOW5.IO3 Improve management HLO22.AOW5.IO5 Exchange genetics	breeding network performance. There may be a combination of centralization and standardized tools or processes to address bottlenecks. There is feedback gained through monitoring of breeding pipelines and performance assessment that informs continuous improvement.	throughout breeding activities and adjunctive, impact enabling work. The approaches used to resolve bottlenecks champion CGIAR values and policies such as FAIR principles, gender equality and transdisciplinary innovation.	challenges or increase return (impact) on investment? (AOW5 E)	Organizational change is needed to overcome bottlenecks. There is enough commonality in organizational challenges that common solutions can be effectively applied.

Research questions and their expected use

Table 4 Expected utilization of MELIA findings

KEQ	Evaluation criteria	Research Questions (RQ)	Utilization
KEQ1. How has B4T's contribution to the impact areas (return on investment) been maximized?	Relevance	RQ1 To what extent can impact be maximized, without variety dominance that reduces system diversity and resilience? (AOW1 MI) RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products? (AOW2 AB)?	Determine appropriate targets that balance trade-offs between widespread adoption with dominant varieties and resilient, diversified systems. This may shift thinking from maximized impact to optimized impact, or assist with developing packages of products for market segments that collectively maximize impact. Inform how much resource reallocation is required to deliver products. Over reallocation can stifle impact just as much as under resourcing.
	Sustainability	RQ19 How has the changed approach to networked partnerships contributed to more sustainable breeding networks? (AOW5 E) RQ20 How can long-term replacement strategy planning support scaling partners to prepare for optimum lifespans of developed varieties? (AOW3 ID)	Inform strategies that can encourage development of the network to have less reliance on CGIAR centers, such as in routine breeding. Develop understanding on limits to impact. Climate change requires ongoing variety development, meaning the variety development pipeline requires sustainability, as opposed to the products. Products need to have limited legacy, with turnover every 10 years for climate adaptation.
	Impact	RQ13 Have the expected projections from product concept and design translated into on ground results, and if not, why not (AOW2 AB)? RQ14 What additional impact could be achieved with additional investment? (AOW1 MI) RQ15 What is the trajectory of genetic gain for each crop, and what benefits could this mean for smallholder farmers who adopt improved products? (AOW4 BR) RQ16 When is the scale of adoption for recently released varieties and products in the pipeline expected to be sufficient to reduce weighted average variety age (WAVA) in targeted market segments to 10 years or less? (AOW3 ID)	Demonstrate impacts achieved by B4T and encourage ongoing investment Identify opportunities to improve uptake and adoption, to maximize impact Guide future target setting

		RQ17 What proportion of marginalized beneficiaries in each market segment have accessed improved varieties? (AOW3 ID) RQ29 What extent of value proposition is needed to motivate adoption?	
KEQ2. To what extent has B4T and its AOW effectively achieved the intended intermediate outcomes?	Coherence	RQ4 How have the network strategies improved coherence with region x crop investments? (AOW5 E) RQ5 How have bottlenecks or barriers between variety development and seed systems been overcome to improve smallholder access to seed? (AOW3 ID) RQ6 How has B4T's work with other CGIAR investments increased or improved the achievement of 2030 outcomes? (AOW5 E)	Maximize leveraging of other investments to increase impacts from B4T investments
	Sustainability	RQ18 To what extent have capacity improvements been maintained within breeding networks? (AOW5 E)	The central coordination of capacity building offers the opportunity to get more strategic, with more sophisticated longer-term capacity planning. This can align with and enable network goals. This new approach requires learning and continuous improvement.
	Efficiency	RQ10 To what extent have systemic, organizational changes improved breeding efficiency and effectiveness? (AOW5 E) RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR)	While these RQ address efficiency, the findings should also inform how these changes and approaches are improving the CGIAR-NARES competitive advantage.
	Quality of Science	RQ21 To what extent have breeding targets been successfully achieved? (AOW2 AB) RQ22 How has the scaled use of modernized practices contributed to maintaining CGIAR's competitive advantage in breeding? (AOW4 BR) RQ23 What contribution have game-changing genetic resources made to impact from variety development? (AOW2 AB)	Demonstration of abilities and successful product development to encourage additional funding. Build investor confidence
	Effectiveness	RQ2 How did prioritization using the Design Standard contribute to maximized impact? (AOW1 MI)	Check if the processes designed to maximize impact are working and identify any areas for improvement. Holistic understanding of collaboration results and requirements, moving collaboration on from

		RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products? (AOW2 AB)? RQ7 To what extent has B4T increased uptake of varieties by scaling partners who have a widespread reach (collectively)? (AOW3 ID) RQ8 To what extent has CGIAR-NARES prioritization of products encouraged targeted investment for breeding? (AOW2 AB) RQ9 To what extent has the amount and quality of active collaboration on products increased? (AOW5 E) RQ11 To what extent has B4T influenced more equitable access to improved varieties by marginalized groups? (AOW3 ID) RQ27 To what extent have interoperable tools and services enabled new and improved collaborations? (AOW4 BR)	coordination around common activities/goals to shared decisions and shared data
KEQ3: How can AOWs influence on intermediate outcomes be increased?	Learning	All research questions listed under KEQ2 plus, RQ24 Have the AOWs effectively come together in transdisciplinary collaboration that holistically encompasses breeding from market targeting through to delivery to smallholders? (AOW5 E) RQ25 What organizational changes are needed to overcome challenges or increase return (impact) on investment? (AOW5 E)	Inform continuous improvement and design of future investments

B4T Program Logic research questions, and indicators

Table 5 Results framework, with applied B4T indicators

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
2030-OC1 Farmers' widespread use of significantly improved products results in a transformative scale of benefits for farmers, consumers and society in the Global South	12.1 Number of food producers using CGIAR innovations 8.3 Realized genetic gains in farmer-relevant conditions 9.1 Weighted average age of varieties in farmers' fields for a prioritized set of representative crops and countries	Improved varieties provide a large-enough gain in value proposition for smallholder farmers that it motivates individual farmers to replace varieties that are over 10 years old. <i>Note HLO4.AOW1 Foster motivations will quantify 'large-enough'. (RQ 29)</i> Provisionally, a productivity gain of 30% is considered transformational for an individual farmer. There is a large number of farmers taking up improved products, with adoption replacing the market share of older varieties. Provisionally, a market share between 25% and 50% is a large enough proportion to have significant impact, while ensuring some variety diversity and resilience remains. <i>Note HLO2.AOW1. Target markets will need to determine optimal levels of variety uptake. (RQ1)</i> The extent of improvement in the variety, combined with the number of farmers using the variety results in a large total benefit in targeted market	RQ13 Have the expected projections from product concept and design translated into on ground results, and if not, why not (AOW2 AB)?	Product Performance Results Table	13.1 OFVT performance results relative to the defined TPP targets, summarized as average percentage of target achievement for all improve traits (GF 6.1)
			RQ14 What additional impact could be achieved with additional investment? (AOW1 MI)	Investment Cases	14.1 Projected number of people with increased access to safe and nutritious food due to additional investment 14.2 Projected number of people living in rural areas lifted to above the extreme poverty line of US \$1.90 per day due to additional investment
			RQ15 What is the trajectory of genetic gain for each crop, and what benefits could this mean for smallholder farmers who adopt improved products? (AOW4 BR)	Crop Performance Reports	15.1 Predicted and realized rate of genetic gain (late-stage, on station) (GF 5.1)
			RQ16 When is the scale of adoption for recently released varieties and products in the pipeline expected to be sufficient to reduce weighted average variety age	Track	16.1 WAVA projection per targeted/prioritized market segment 16.2 EGS or certified seed production 1-3 years after release (GF 2.1)

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
		segments that is detectable in a correlating shift in trajectory of targeted global measures in the market segment.	(WAVA) in targeted market segments to 10 years or less? (AOW4 ID)		16.3 Prevalence of breakthrough product on landscape 3-5 yrs after release (or after made available to certified system) (GF 2.2)
			RQ6 How has B4T's work with other CGIAR investments increased or improved the achievement of 2030 outcomes? (AOW5 E)	To be determined/ developed	6.1 Number of farmers adopting improved varieties with direct contributions from other CGIAR investments? 6.2 Efficiency gains through collaboration with Digital Accelerator
2030-OC2 Women, youth and other marginalized groups in food, land and water systems are empowered by and benefit from improved products in the Global South	12.1 Number of food producers using CGIAR innovations, disaggregated by gender	A large proportion of marginalized groups gain access to seeds of improved varieties. The use of these varieties, and benefits that ensue, are sufficient to empower marginalized groups and reduce their level of disadvantage. The proportion of marginalized product users in each market segment is known, and (provisionally) over 50% of these users access varieties that are designed specifically for them.	RQ17 What proportion of marginalized beneficiaries in each targeted market segment have accessed improved varieties? (AOW3 ID)	Track	17.1 Percentage of marginalized groups accessing improved varieties within 5 years of variety release
IO1 CGIAR-NARES prioritization of transdisciplinary designed high impact products encourages more targeted breeding by networks	3.2 Number of research partners from the Global South that collaborate with CGIAR 4.2 Value (in USD equivalent) of	The breeding network competently uses social science data to inform product prioritization, and this use of social science data makes the prioritization process transdisciplinary. The inclusion of social science in the Product	RQ8 To what extent has CGIAR-NARES prioritization of products encouraged targeted investment for breeding? (AOW2 AB)	PMP	8.1 Percentage of CGIAR-NARES funding reallocated to prioritized products 8.2 Average age of parents in new crosses (GF 5.2)

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
	public investment by national governments in policies and programs informed by CGIAR research	Design Standard is sufficient to ensure decision-making overcomes existing breeding biases. The prioritized products aim to deliver (provisionally) at least 80% of the averaged projected benefits across the product concepts used to inform product designs.			
HLO7.AOW2.IO1. Accelerating genetic gain	Other outputs – Average age of parents in crosses 8.2 Predicted genetic gains in breeding programs 8.1 Proportion of pipelines that have implemented systems to measure predicted and realized genetic gains on farm	Across the CGIAR-NARES breeding network resources are reallocated to agreed priorities, while maintaining distributed ownership and empowerment across the network. Partners are committed to the prioritized products. The extent of reallocation is sufficient to ensure prioritized products are successfully developed within the shortest possible timeframe. <i>Note, the extent of reallocation or sufficient level of investment for successful product development needs to be determined.</i>			
IO2 CGIAR Centers, AOWs and NARES actively collaborate on successful product development	7.2 Proportion of CGIAR-NARS-SME breeding pipelines within CGIAR-NARS-SME network using market segments, target product	Active collaboration is where knowledge sharing prompts discussion in a process of curiosity and discovery. This contrasts with passive collaboration where knowledge is provided or gained, without coming together to explore	See RQs assigned to contributing HLOs below	Product Performance Reports	

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
	profiles to guide selection decisions for all advancement steps 7.1 Proportion of new released varieties developed in alignment with market intelligence-informed target product profiles and tailored to market segments in the Global South	differing experiences or perspectives. Active collaboration occurs across organizations, scales and work units.			
HLO9.AOW2.IO2 Transform collaboration	Other outputs - Number of partners involved in breeding decisions		RQ9 To what extent has the amount and quality of active collaboration on products increased? (AOW5 E)	Breeding Portal	9.1 Number of partner institutions that provided usable testing data to network in each stage of testing (GF 3.1) 9.2 Only relevant for subset of networks: Proportion of stage three entries that originate from NARES crosses (GF 3.4)
			RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products? (AOW2 AB)?	To be determined/ developed	3.1 Distribution of funding in network between partners (CGIAR Centers: NARES) (GF 3.5) 3.2 Percentage of funding allocated to prioritized products

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
			RQ27 To what extent have interoperable tools and services enabled new and improved collaborations? (AOW4 BR)	Breeding Portal	27.1 Number of data analyses conducted that use data that is gathered across more than one organization
HLO10.AOW2.IO2 Create products	Number of innovations (innovation development) – number of varieties released to partners for scaling	Successful product development means that the scale of improvement targeted, as defined to deliver transformative impact, is achieved.	RQ3 How has resource reallocation (due to prioritization) contributed to successful development of high-impact products? (AOW2 AB)?	PMP/Breeding Portal	3.3 OFVT performance results table on key adoption traits plus yield, relative to the defined TPP targets and replacement variety values (GF 6.1) 3.4 Stage 1-4 performance results table (GF 5.3)
IO3 CGIAR-NARES breeding network cost-effectively develops prioritized products, in the shortest possible time	3.1 Number of national research institutions and analytical units in the Global South using new or improved datasets from CGIAR	Breeding efficiency is more than just the cost of breeding. Efficiency is the cost of successful product development (i.e. cost-effectiveness). This interpretation combines efficiency from IO3 with successful in IO2, ensuring compatibility of these outcomes. Ensuring successful breeding may require increased cost where previous investment has been insufficient. Products prioritized based on impact may be more expensive to develop. The shortest possible time varies between crops due to the biological growth cycle.	RQ10 To what extent have systemic, organizational changes improved breeding efficiency and effectiveness? (AOW5 E)	To be determined/ developed	10.1 Budget allocated for each targeted product 10.2 Cost of delivery for each targeted product
			RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR)	Breeding Portal	12.1 Expenditure by breeding programs to use shared services 12.2 Estimated expenditure for comparable use via individual arrangements 12.3 Number of organizations accessing shared tools and services that would not have accessed comparable tools and services independently

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
IO4 NARES and SME scaling partners increase uptake of varieties with step-change benefits	Number of CGIAR-supported innovations being used by next users	The market share of engaged scaling partners collectively covers a large enough portion of the targeted farmers to allow for below 100% adoption rates and still meet impact targets. Scaling partners have the information and resources required to position products to the markets and gain sufficient returns to motivate changes to seed supply.	RQ7 To what extent has B4T increased uptake of varieties by scaling partners who have a widespread reach (collectively)? (AOW3 ID)	Track	7.1 Number of scaling partners taking up improved varieties 7.2 Percentage of seed market covered by engaged scaling partners
IO5 NARES and SMEs promote equitable access to improved varieties, by marginalized groups	Quantity of seed produced and certified for a prioritized set of representative crops and countries	Equitable access is fair and equal access to improved varieties, ensuring that marginalized groups are not further disadvantaged and not left behind as others take up improved varieties. Fair and equal access is measured proportionately with an equal proportion of marginalized groups in the target market segment access varieties as the proportion of non-marginalized users of improved varieties.	RQ11 To what extent has B4T influenced more equitable access to improved varieties by marginalized groups? (AOW3 ID)	Track	11.1 Quantity of seed produced and certified for varieties released within the last 5 years
R1.AOW1 Knowledge developed by Market Intelligence is available to breeding networks to inform impact- and demand-		The use of market intelligence data for decision-making requires three conditions to be met: The quality and completeness of the information available			

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
oriented product concepts.		- Accessibility of the information to decision-makers - Application and interpretation of the information within decision-making processes Each of these three conditions needs to be separately monitored for insights on improvement points along the process. The quality and completeness of the information can be monitored internally by Market Intelligence. Accessibility can be inferred from number of times decision-tools are accessed, and the number of unique users (IP addresses) and repeat users. The last two conditions will benefit from some level of feedback from data users.			
HLO1.AOW.IO1. Steer to impact	Number of knowledge products 5.2 Number of multi-stakeholder platforms taking system-related decisions or making recommendations based on CGIAR technical support 3.1 Number of national research institutions and analytical units in the Global South using new or improved datasets by CGIAR		RQ28 To what extent is market intelligence used?	GloMIP metrics	28.1 Number of times decision-tools are accessed 28.2 Number of unique users 28.3 Number of repeat users
HLO2.AOW1.IO1 Target markets	Number of knowledge products		RQ1 To what extent can impact be maximized, without variety dominance that reduces system diversity and resilience? (AOW1 MI)	Projection of benefits	1.1 Optimal impact targets in product concepts 1.2 Average number of unique potential beneficiaries for each target market
HLO3.AOW1.IO1 Design concepts	Number of knowledge products	Standard and prioritization processes. A comparison of potential (pre decision) and forecast impact (post decision) will be informative of the	RQ2 How did prioritization using the Design Standard contribute to maximized impact (AOW1 MI)?	Product Design Standard Reporting	2.1. Impact targets of prioritized products 2.2 Average projected benefits per product concept

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
		interpretation of data and highlight other factors that may be coming into play to limit impacts from breeding.			2.3 Total hectares of market segments with current product concepts
HLO4.AOW1.IO1. Foster motivations	Number of knowledge products		RQ29 What extent of value proposition is needed to motivate adoption?	Documentation of behavioral science	29.1 Number of target markets with behavioral science insights
R2.AOW2 Feasible strategies for target product profiles and pipelines are developed		Feasible means achievable within currently available technologies and tools. The resourcing required to deliver each feasible product profile and pipeline is estimated to ensure the next stage of prioritization for resource allocation results in sufficient condensing of resources to achieve required technical results.	RQ21 To what extent have market-set targets been successfully converted into breeding targets? (AOW2 AB)	Product Design Standard Report	21.1 Percentage of projected impact from product concepts converted into breeding targets
HLO6.AOW2.IO1. Refocus to markets	Other outputs – number of breeding pipelines	Quantitative targets for the level of improvement required for all value propositions are set. The targets are sufficient to motivate uptake by scaling partners and adoption by smallholder farmers, as informed by market intelligence's data.			
R3.AOW2 Traits and genetic resources with significant potential for genetic gain are developed by Accelerated	Number of innovations (innovation development)	The resources are available to breeders and used to develop improved varieties. This requires breeders to have the capacity required to make use of the resources and ability to access them.	RQ23 What contribution have game-changing genetic resources made to impact from variety development? (AOW2 AB)	BPAT	23.1 Updated inventory of high value haplotypes agreed on by network for deployment for the MS/TPP (including their readiness levels and effect estimates) (GF 4.1) 23.2 Frequency of high value haplotypes in new crosses, and

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
Breeding, in collaboration with ARIs and NARES		A step-change level of improvement is (provisionally) a 50% gain or greater in performance of the targeted trait in haplotypes. This level of improvement allows for some loss as traits are brought into existing lines. Collaboration with ARIs and NARES is more than input into decision-making. It includes shared resources, including genetics and constructive joint-problem solving as difficult breeding challenges are embraced and innovations attempted.			each testing stage thereafter, as well as releases (GF 4.2)
HLO8.AOW2.IO2. Discover genetic potential					
R4.AOW5 Multi-scaled, multi-disciplinary networked partnerships coordinate capacity development, enabling modernized breeding		Networked partnerships are dynamic and complex, with several interrelationships. There are a greater number of connections that do not flow through CGIAR centers. Breeding priorities and efforts remain collaborative towards common goals. Network strategies are comprehensive and set the long-term vision for the network, with clear links to the Intermediate Outcomes and Impacts. Network strategies enable forward planning for all partners, with links to replacement strategies so			
HLO18.AOW5.IO2 Partnered networks	Other outputs - number of breeding network strategies		RQ4 How have the network strategies improved coherence with region x crop investments? (AOW5 E)	Partnered Network Survey	4.1 Number of partners with national strategies that align with network strategy objectives and breeding targets 4.2 Perceived coherence by network partners

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
		that future capacity requirements can be developed to ensure future breeding priorities can be delivered. Connections between breeding networks enable greater spillover effects, with innovations developed shared across networks. There may be common links or bridges across networks, with networks operating across geographic scales.	RQ19 How has the changed approach to networked partnerships contributed to more sustainable breeding networks? (AOW5 E)	Partnered Network Survey	19.1 Number of partners accessing support (financial, expertise) from non-CGIAR partners within the network
HLO19.AOW5.IO2 Share capacity	Number of people trained (capacity sharing for development) 3.6 Number of individuals from research or scaling partner organizations who attended CGIAR trainings		RQ18 To what extent have capacity improvements been maintained within breeding networks? (AOW5 E)	Breeding Station Survey	18.1 Percentage of staff with sufficient level of self-rated knowledge and skills
R5.AOW4 Interoperable breeding tools and services that are tailored to the capacity of partners are used in collaborative efforts across the breeding network	Other outputs – number of tools and services used 5.2 Number of multi-stakeholder platforms taking system-related decisions or making recommendations based on CGIAR technical support	A hub and spoke model of scaled tools and services allows partners of varied capacity and resourcing to make valuable contributions to shared breeding goals. The design of scaled tools and services empowers hubs to work directly with spokes, establishing a sustainable model of respectful, regionalized capacity building.	RQ22 How has the scaled use of modernized practices contributed to maintaining CGIAR's competitive advantage in breeding? (AOW4 BR)	Breeding Portal	22.1 Number of organizations making use of scaled tools and services
HLO15.AOW4.IO2 Hub & spoke services				Partnered Network Survey	22.2 Improvement in relationships due to hub & spoke scaled tools and services
R6.AOW4 The CGIAR and NARES breeding network	Other outputs – number of tools and services used	A critical mass of tool and service users ensures that a minimum operating volume of use is met.			

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
access cost-effective breeding tools and services	5.2 Number of multi-stakeholder platforms taking system-related decisions or making recommendations based on CGIAR technical support	The volume of users maintains and improves negotiating power with commercial providers of tools and services. The quality of the tools and services developed by CGIAR are attractive to commercial breeders, generating interest and additional users.			
HLO16.AOW4.IO3 Shared services			RQ12 How has the use of interoperable, scaled breeding tools and services increased the cost-effectiveness of the network? (AOW4 BR)	Breeding Portal	12.1 Estimated cost-savings from the use of scaled tools and services, as opposed to using non-scaled tools and services
HLO17.AOW4.IO3 FAIR principles	Other outputs - Number of service users complying with FAIR	Modernized breeding practices (e.g. quantitative genetics) that accelerate and improve genetic gain are routinely used across the majority of the breeding pipelines.			
R7.AOW3 Scaling partners pursue opportunities to replace existing lesser performing varieties with improved varieties		All policy and institutional barriers have been overcome. The only remaining barrier to the uptake of improved varieties by scaling partners is insufficient expected benefits or returns of improved varieties. This is the			
HLO11.AOW3.IO4 Position for demand	Number of knowledge products – strategies published 2.3 # of private sector actors using CGIAR innovations 1.1 Number of CGIAR-supported	returns from the perspective of the scaling partner i.e. the market forces that flow through from smallholders due to performance gains of improved varieties relative to any scaling costs of changing varieties. Scaling partners have improved understanding of the benefits of changing varieties and are motivated to make use of improved varieties.	RQ20 How can long-term replacement strategy planning support scaling partners to prepare for optimum lifespans of developed varieties? (AOW3 ID)	Product advancement reports	20.1 Projected seed production of advancing products 20.2 Forecast returns for scaling partners that take up new varieties 20.3 Registration ambition and sequence by network member country for breakthrough products (GF 8.2) 20.4 End-user preference ratings of candidate line relative to replacement (GF 6.2)

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
	innovations being used by next users				
R8.AOW3 The supply chain rapidly disseminates improved varieties at scale		Adoption of improved varieties is not limited by farmer access to seed. There is a small surplus volume of seed available each year for varieties that are less than 10 years old. (A consistent approach for measuring age of variety is under development. This is expected to take into account delays between CGIAR development and release by national partners).			
HLO12.AOW3.IO5 Deploy seed	Number of knowledge products – strategies published 1.3 Quantity of seed produced and certified for a prioritized set of representative crops and countries		RQ5 How have bottlenecks or barriers between variety development and seed systems been overcome to improve smallholder access to seed? (AOW3 ID)	Track	5.1 EGS or certified seed production 1-3 years after release (GF 7.1) 5.2 Percentage of released varieties in the past 5 years that are in commercial production (GF 8.1)
HLO13.AOW3.IO5 Power seed scaling	Number of knowledge products – strategies published 1.3 Quantity of seed produced and certified for a prioritized set of representative crops and countries				
HLO14.AOW3.IO5 Track	Number of knowledge		RQ16 When is the scale of adoption for products	Track	16.1 Prevalence of breakthrough product on

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
	products – annual publication of tracking results		in the pipeline expected to be sufficient to reduce weighted average variety age (WAVA) in targeted market segments to 10 years or less? (AOW4 ID)		landscape 3-5 yrs after release (or after made available to certified system) (GF 7.2) 16.1 WAVA (GF 7.3)
R9.AOW1 Impact evaluation is informed by Market Intelligence's analysis of potential benefits of investment options	Number of knowledge products	The information in these investment cases can be used as a check on the level of impact currently being delivered, and discrepancies can be explored to improve either breeding processes or projections. Monitoring and evaluation provide evidence for assumptions in projection models, increasing confidence in the projections of impact as a reliable impact data point. Projections of benefit can be readily updated as products progress through the breeding pipeline and finally based on results of on-farm trials.	RQ14 What additional impact could be achieved with additional investment? (AOW1 MI)	Investment cases	14.1 Forecast Return (impact) of Investment of breeding pipelines (return is defined as additional impacts compared to current benchmark varieties) (GF 1) 14.2 Number of investment cases developed
HLO5.AOW1.IO3. Investment cases					
R10.AOW5 CGIAR and NARES identifies and addresses systemic organizational bottlenecks to improve breeding		Organizational change supports the widespread adoption of modernized breeding, with a high quality of science and cost-effectiveness. Cultural changes continue with continuous improvement embedded throughout breeding			

Outcome/Output	CGIAR Indicator	Definitions of success	Research Questions (RQ)	Measurement tool	B4T indicators
network performance		activities and adjunctive, impact enabling work.			
HLO20.AOW5.IO3 Assess performance	Number of knowledge products – number of MELIA studies completed	The approaches used to resolve bottlenecks champion CGIAR values and policies such as FAIR principles, gender equality and transdisciplinary innovation.	RQ25 What organizational changes are needed to overcome challenges or increase return (impact) on investment? (AOW5 E)	To be determined/ developed	25.1 Descriptive results
HLO21.AOW5.IO3 Improve management	Other outputs - number of strategies, policies and SOPs developed		RQ24 Have the AOWs effectively come together in transdisciplinary collaboration that holistically encompasses breeding from market targeting through to delivery to smallholders? (AOW5 E)	Partnered Network Survey	24.1 Distributed costs across OFVT (GF 9.4) 24.2 Satisfaction level with contribution from each AOW
HLO22.AOW5.IO5 Exchange genetics	Other outputs - number of genetic products shared 11.1 # Genetic resources exchanged in compliance with phytosanitary regulations		RQ9 To what extent has the amount and quality of active collaboration on products increased? (AOW5 E)	PMP	9.3 Number of partners contributing lines or clones for shared population development (GF 3.2)

5. Future work required to complete MELIA planning

In the timeframe available to develop this strategic MELIA framework, it has not been possible to finalize the MELIA plan requirements. Effort has been invested to set a strong foundation for more detailed MELIA planning. This has meant getting a quality structure in place, with a TOC, logical framing of research questions, and foundational indicators that will be informative and deliver on the MELIA objectives for B4T and associated prioritized evaluation criteria.

Now that this strategic level of planning has been completed, it is possible to define more operational level details such as relevant indicators and the tools required to ensure required data is collected in an efficient manner. Remaining work includes:

- A monitoring plan for HLOs and outcomes, including a data collection strategy that will efficiently deliver on the data and evidence needs of any planned evaluations and impact assessments.
- An evaluation plan with identified evaluation studies and evaluation collaborations that meet the B4T MELIA objectives.
- An impact assessment plan with utilitarian emphasis to contribute to the B4T MELIA objectives, in particular continuous improvement.
- Learning and adaptive management, including annual review and planning to respond to learnings as emerging.
- Reporting processes with assigned responsibilities.
- Assignment of MELIA resources, with budget and staff to implement the MELIA plan
- Data management that ensures compliance with CGIAR's Open and FAIR Data Assets Policy.

These next steps will cover headings 3 to 9 of the MELIA plan guidance documentation. At this point in time, it is expected that the evaluability assessment for domains A and B will only be partially possible, but that the quality of what is provided and the rigorous basis it provides will be recognized.

Table 6 Plan of actions to complete operational level MELIA planning

Activity	Tasks	Timeline
A monitoring plan for HLOs and outcomes, including a data collection strategy that will efficiently deliver on the data and evidence needs of any planned evaluations and impact assessments	Fill indicator gaps to ensure high-level indicators for each HLO that can evidence responses to Research Questions Identify indicators for Intermediate Outcomes Allocate indicators to existing monitoring tools	End of June
	Work with each AOW to ensure indicators (qualitative and quantitative) are incorporated and tools will produce evidence against Research Questions Design monitoring tools for any remaining gaps and allocate to AOWs Identify if additional resourcing is required Complete data collection strategy for each monitoring tool Commence MELIA implementation with baseline data collection Independently review baseline data for sufficiency and refine planning accordingly	End of August
An evaluation plan with identified evaluation studies and evaluation collaborations that	Determine appropriate scale of evaluation studies with input from AOW leads. There may be a small number of keystone evaluations on strategically important elements e.g. contribution of prioritization to impact, gains and areas of improvement for networked partnerships	End of August

meet the B4T MELIA objectives	Identify evaluation studies required to bring together multiple lines of evidence, and required timing for these studies based on how the findings can be applied to improve B4T.	
	Scope TOR for each evaluation study and set clear expectations on deliverables Determine capacity needs for each evaluation study Assess benefits of internal evaluation compared to external independent evaluation Identify milestones required to allocate and implement evaluation	End of September
An impact assessment plan with utilitarian emphasis to contribute to the B4T MELIA objectives, in particular continuous improvement	Note, impact is an evaluation criterion. Impact Assessments and RCTs are possible methods to evaluate impact. Other methods are available to determine impact. The most appropriate method that provides useful information to improve B4T needs to be determined. E.g. Impact assessment can be used to increase confidence in projection of benefit assumptions. A detailed review of the in-progress Impact Assessment will be required to determine potential use of findings beyond accountability.	End of September
Learning and adaptive management, including annual review and planning to respond to learnings as emerging	Collate list of indicators and research questions that will have data available on an annual basis Review breeding pipeline performance management in Enable for MELIA alignment	End of June
	Develop discursive process to facilitate internal learning and empower the breeding system to identify recommendations improvements	End of July
	Determine suitable timeframe for external review of breeding system recommendations for improvements, and potential to incorporate into existing MELIA tools such as BPATs (to reduce risk of entrenched bias and group think, adding rigor to recommendations). Develop process to allocate actions across AOW and assess potential for resource reallocation to enable actions. Develop process for tracking and reporting of implemented improvements.	End of September
Reporting processes with assigned responsibilities	Align reporting process and data collection strategy Assign responsibilities, including oversight	End August
Assignment of MELIA resources, with budget and staff to implement the MELIA plan	Collate existing resourcing for AOW embedded MELIA activities Check sufficiency of resourcing with AOW leads Identify any additional monitoring resourcing required Identify any additional resourcing required for evaluation studies	End October
Data management that ensures compliance with CGIAR's Open and FAIR Data Assets Policy	Quality check the data collection strategies for compliance	End of August

6. Appendices

Feedback on proposal TOCs and recommendations from the Genetic Innovation Science Group evaluation

Constructive feedback received from the ISDC covered the TOC and/or MELIA. These include:

- *'The overall Theory of Change illustration (Fig 1) lacks alignment with the 22 high-level outputs (Table 1) and needs better synchronization to be credible.'*
- *'The details are not clear. For instance, the links between outputs, outcomes, and impacts are not clear, with the large arrows from "Areas of Work" to "High-Level Outputs" and "2030 Outcomes" to "Impact Areas" being too unspecific.'*
- *'More information could be given regarding the feedback loops among the various Areas of Work. For example, How MARKET INTELLIGENCE, ACCELERATED BREEDING, INCLUSIVE DELIVERY and BREEDING RESOURCES are related to learning opportunities.'*

The TOC showed only 16 out of 22 tabulated HLOs with overlapping and merging lines from these to the intermediate outcomes, working to obscure relationships and accountabilities. The link between AOW and HLOs was also not evident. There was little clarity, in either the B4T TOC or underpinning AOWs, on the causal pathways from HLOs to intermediate outcomes. In response,

- All HLOs are now visible in the B4T TOC.
- AOW results have been included to show causal relationships from HLOs to intermediate outcomes.
- The intermediate and 2030 outcomes are consistent across the AOW TOCs.
- AOW TOCs are now fully nested within the B4T TOC, clearly showing contribution from HLO, to AOW to B4T's intermediate and 2030 outcomes.

It is noted that points of feedback not specific to the TOC need to be incorporated into the TOC and MELIA plan. Examples include feedback on the integration of feedback mechanisms, defined strategies to enhance stakeholder engagement and co-creation, prioritization to market driven products along with equitable access for marginalized groups, and partnerships across the breeding network and system. These non-product paths to impact noted in the feedback have been considered in the drafting of outcomes to ensure the approach taken by B4T is in alignment with ISDC directions and the Ways of Working.

The issues raised in the ISDC feedback are evidently longer-term issues in the recognition, adoption and application of evaluation science in breeding endeavors. While an earlier TOC showed the relationship between the outcomes of the initiatives, the revised TOC showed only one outcome of breeding. This outcome was 'Seed producers successfully market new demand-driven improved varieties to more farmers'. The GI TOC essentially elevated the final output of the breeding pipeline to an outcome³, reverting to the breeding pipeline engineering view that dominates in breeding. In elevating the output to an outcome, any outcomes outside of breeding itself are diminished. Outcomes related to the quality of how breeding is done, with strategic prioritization, collaborative partnerships and efficiency have been omitted. It is these qualities of how breeding is done that build empowerment of regional partners and legacy beyond the lifecycle of a variety.

³ Note that it could be argued this is an outcome of Seed Equal. Regardless, it is not an outcome of GI, but one of the initiatives and this outcome could be achieved without investment or action in any other initiative.

In the absence of embracing social science, and in particular evaluation science, to demonstrate impact to Funders and other stakeholders, B4T is reliant on trust, reputation and historically established competitive advantage to justify its ongoing investment.

The vision for MELIA of B4T is for an integrated framework that connects across the breeding pipeline, from the economic science of understanding market needs to the genetic science of breeding and the social and policy science of seed delivery systems, with evaluation science demonstrating how these parts deliver worthwhile impact. This requires a robust and logical Theory Of Change (TOC) that connects the outcomes of breeding that occur across the pipeline and transdisciplinary areas of work required to maximize the impacts achieved through improved varieties. The TOC is pivotal to mapping out the complex connections across a holistic B4T program and enabling clarity in responsibilities for MELIA and accountabilities for outcomes.

ISDC Feedback

The ISDC Feedback was summarized for the System Council. This summarized feedback is shown in Table 7, along with how the points of feedback have been considered in the revision of the TOC and MELIA framework.

Table 7 Response to ISDC Feedback from MELIA perspective

Assessment Criteria	Comment	Response
2. Evidence that the Science Program is demand driven through codesign with key stakeholders and partners (NARES, governments, farmers, private sector, funders) and research collaborators within and outside CGIAR.	This Program targets “impactful geographics and user needs,” seeking to identify areas with the lowest 35% impact and reallocate these resources to the top 25% impact. However, more acknowledgement could be given to the crucial role of the NARES and farmers and their local knowledge which should be integrated with modern tools to enhance adoption. While targeting “the highest scale for potential impact,” care must be taken to include the most vulnerable and poor. More detail could be given to the methods of the implementation of “co-creation and co-testing.”	The MELIA framework has identified the need to determine appropriate resource reallocations. The assumption that a reallocation based on lowest 35% impact results in sufficient resourcing for success requires testing.
4. Research questions address well defined knowledge gaps and emerging megatrends, with a particular emphasis on climate change, and are supported by underlying hypotheses.	The research questions and hypotheses are clearly stated but could be improved by being more quantitative and giving clarity to how they can be tested. For instance, research question in the Program-level Theory of Change as shown in fig. 1 seems to be too general and not clearly actionable. For example, “How do breeding network actors innovate and strengthen their research...?” is answered with “Breeding network actors increase uptake of quality products in priority market segments.” In addition, potential negative outcomes and time frames could be discussed.	Research questions have been redesigned to address the Key Evaluation Questions and evaluation criteria. This has ensured all research questions will contribute to assessing current impact from investments and informing continuous improvement. Research questions have also been mapped to the refined TOC. This structured approach has avoided general questions and focused on producing knowledge that is informative for evaluation and can be applied in the future to continuously improve B4T and its impacts. The applied

		breeding indicators for each HLO will gather quantitative data. Qualitative research will be required to assess relationship changes, such as the effectiveness of new partnership approaches.
5. Theory of Change with intended outputs, outcomes, and impacts at scale clearly described. Assumptions are documented, causal linkages are clear, especially the role of partners in driving impact through inclusive innovation, and all indicators made explicit.	The Theory of Change in fig. 1 is effective in discussing the overarching goals of this proposal and the intended effects on society. However, the details are not clear. For instance, the links between outputs, outcomes, and impacts are not clear, with the large arrows from “Areas of Work” to “High-Level Outputs” and “2030 Outcomes” to “Impact Areas” being too unspecific. In addition, integration with other Programs is missing. Fortunately, Table 2 helps to clarify many of these problems as it includes Outcome Statements, Partners, Assumptions, and Indicators. In addition, the individual Area’s of Work Theory of Changes add clarity to the overall Theory of Change, as discussed on pages 60-64.	The TOC has been refined. All HLOs are now shown. As a result, there has also been refinement to the outcomes. The Intermediate Outcomes are less linear. Pathway to impact from the HLOs is now clear. AOW TOCs are now fully nested within the B4T TOC, more clearly showing how each AOW contributes to the whole.
11. MELIA approach that supports effective adaptive management and learning. Lessons are used to proactively reflect on and adapt the Theory of Change. Impact assessment strategy outlined.	The MELIA approach will be managed by dedicated full-time officers and monitored regularly. Data collection is aligned with the key elements of the Theory of Change, which could be improved with better monitoring for other unintended outcomes, political instability, urban framing, displacement and the integration of generative AI tools for monitoring. One problem is that it is not clear at what stage various people will become involved in the process, and those who are supposed to benefit will be associated. An early assessment should be whether the new varieties are appreciated by the target farmers or not. More information could be given regarding the feedback loops among the various Areas of Work. For example, How MARKET INTELLIGENCE, ACCELERATED BREEDING, INCLUSIVE DELIVERY and BREEDING RESOURCES are related to learning opportunities.	A robust framework for MELIA has been developed, with the TOC providing a consistent structure for all MELIA components. This will ensure that MELIA activities undertaken by each AOW will connect to form a comprehensive evaluation of outcomes. The high-level MELIA plan outlines the steps now required to operationalize the framework, embedding it into existing tools such as the Breeding Program Assessment Tool, Breeding Operations Assessment, Breeding Station Staff Survey, On-farm Trial Product Performance Report and Network Performance Review. MELIA activities will be conducted by all AOWs, making use of the most relevant areas of expertise in applied measurement of the holistic variety improvement system. Enable will provide operational oversight and strategic oversight and collation of results across multiple lines of evidence will be the Director’s responsibility.

